Agent mode in the IDE

Plan first, review the diff, stay in charge

Session M2-02 | 3 hours

Agenda

Segment	Time
Four modes	10 min
What agent mode can touch	12 min
Steering	13 min
Agent mode vs CLI vs cloud agent	8 min
Demo	12 min
Pros, cons, mistakes	5 min

Four modes

Mode	What it does	Changes files?
Ask	Answers questions about your code	No
Edit	Edits the files you pick, in one pass	Yes, only those
Agent	Finds files, edits, runs commands, loops	Yes, and runs things
Plan	Researches and writes a plan	No, until you approve

Pick the **smallest mode** that does the job. Current VS Code lists Ask, Agent, and Plan. Check what your picker shows for Edit.

What agent mode does

```
You: prompt
  → Copilot picks a tool and asks the editor to run it
  → the editor runs it and returns the result
  → Copilot reads the result and picks the next tool
  → ... until done, or it asks you
```

A **tool** is an action the editor can run for the model.

Tools

Tool kind	Examples
Read	Open a file, list a folder
Search	Find text or symbols in the workspace
Edit	Change or create files
Terminal	Run `npm test`, `git status`
MCP	Tools from servers you add: issues, databases, docs

Open the **tools picker** in chat to see and switch them off.

Trust: what it can touch

- **Your files.** It edits them directly. Git is your safety net.
- **Your terminal.** Commands run as you. Each one asks first.
- **MCP servers.** They can reach outside systems with your credentials.
- **Text it reads.** A file or web page can contain instructions. Treat it as untrusted.

You are the **review step**. Nothing else is.

Steering

1. **Plan first.** Use Plan mode, or say "plan, don't edit yet".
2. **Be specific.** Goal, files, how to check it.
3. **Watch the tool calls.** Stop it if it wanders.
4. **Review every diff.** Keep or undo per file.
5. **Undo with checkpoints.** Restore chat to an earlier point.
6. **Read terminal commands** before you allow them.

Terminal approval

Agent mode asks before it runs a command. You can:

- Allow once.
- Allow a command prefix for this session, or always.
- Auto-approve chosen commands with `chat.tools.terminal.autoApprove`.

Tip: auto-approve `npm test`. Keep asking for anything that deletes, installs, or pushes.

Menu labels vary by version. Check the current docs.

Agent mode, CLI, cloud agent

	IDE agent mode	Copilot CLI (M2-01)	Cloud agent (M3-02)
Runs	Your machine, in VS Code	Your machine, in a terminal	GitHub, in its own environment
You are	Watching, in the editor	Watching, in the terminal	Away until the PR
Result	Edits in your working tree	Edits in your working tree	A pull request
Best for	Feature work with diff review	Terminal tasks, scripts	Well-defined tasks you can hand off

TODO code action

```
// TODO: Add GET /export that returns all notes as CSV.
```

With the **GitHub Pull Requests** extension, a lightbulb appears. Pick **Delegate to coding agent**. The cloud agent opens a pull request.

Good for **small, well-described** tasks. You review the PR.

Demo

Same prompt, two modes. Then agent mode with a plan.

- **Ask:** explains what to change.
- **Agent:** changes it and runs the tests.

Watch the tool calls, then the diff.

Pros and cons

Pros	Cons
Multi-file changes in one go	Can edit more than you wanted
Runs tests and fixes failures	Long runs are hard to review
Sees your real project	Wrong assumptions hide in big diffs
Easy to undo	Slower than a quick Ask

Common mistakes

- **Vague prompt.** "Add tags" gets guesses. Name the behavior.
- **No plan.** Big tasks drift.
- **Skipping the diff.** Accepting all without reading.
- **Blanket auto-approve.** One bad command costs more than the clicks.
- **Giant tasks.** Split them.
- **No tests.** The agent has nothing to check against.

Lab

Add a feature with agent mode

Plan, build, review, fix, delegate.